Rotaract Club of Xiamen University Malaysia

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: Not stated in the supplied files — verify the official event notice. DRAFT: team confirmation required before submission.

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	The Powerpuff Girls
Project / Solution Name	BillGuard
Name (Leader)	Aisuluu Abdullaeva
Email & Phone (Leader)	+996552577577 & CME2509072@xmu.edu.my
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://invoice-assistant-workflow-fixes-and-demo.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/alananduya-debug/Powerpuff-Girls

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

1. User interface-> Staff, accountant, and owner views are rendered by React, with the selected userID passed via X-Demo-User-Id header. The frontend uses React Query hooks and Zod-derived request/response types to call the API.
2. API and simulated workspace-> Express receives requests at /api, validates uploads, and resolves the demo identities against a local mock hub containing three users and five supplier records.
3. Invoice reading-> Uses cached fixtures or live OCR (PDF.js, canvas rendering, and image handling) sent to Runpod for JSON extraction, falling back to cached fixtures on failure.
4. Validation and workflow-> The server checks field confidence, required fields, dates, arithmetic, tax rates, and duplicates, moving records through draft, review, fixes, approval, or rejection.
5. Persistence and responses-> Sensitive data and file bytes are encrypted with AES-256-GCM in SQLite, utilizing an HMAC-SHA256 blind index for duplicates. Authorized requests decrypt data on-demand with logging.
6. Hosting-> Configured for a Replit autoscale deployment with a static Vite SPA frontend and a Node.js API service (/api/healthz), relying on local filesystem storage.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3	Frontend, API, shared schemas and tooling.	Apache-2.0
JavaScript (ES modules); Node.js 24	OCR logic and server runtime; demo launch scripts.	JavaScript: N/A (language); Node.js: MIT (with bundled third-party notices)
React / React DOM 19.1.4	Billguard web user interface.	MIT
Vite 7.3.7; @vitejs/plugin-react 5.2.0	Frontend dev server and production build.	MIT

Name & Version	Purpose in Project	License
Tailwind CSS 4.3.3; @tailwindcss/vite 4.3.3	Utility styling and Vite integration.	MIT
Express 5.3.0	HTTP API for invoice upload, review, records and workflow.	MIT
better-sqlite3 13.0.3; SQLite	Local persistence for invoices, line items, events and settings.	better-sqlite3: MIT; SQLite: public domain
Multer 2.5.0	Multipart invoice-file upload handling.	MIT
pdfjs-dist 6.4.299 (PDF.js)	Extracts PDF text and page content for live OCR.	Apache-2.0
@napi-rs/canvas 1.0.13	Renders PDF pages and converts image input for OCR.	MIT package license; review bundled native notices
OpenAI Node SDK 6.49.0	OpenAI-compatible chat-completions client used by the Replit AI integration.	MIT
Zod 3.25.76	Runtime request/response schema validation.	MIT
@tanstack/react-query 5.104.1	Frontend API data fetching and cache/state handling.	MIT
Wouter 3.13.0	Lightweight client-side routing.	MIT
Radix UI React primitives (selected packages; versions vary by component in pnpm-lock.yaml)	Accessible UI primitives used by the interface.	MIT
Lucide React 0.545.0	Interface icons.	ISC
Pino 9.14.0; pino-http 10.5.0	Structured API request and application logging.	MIT
cors 2.8.6	API cross-origin policy middleware.	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Replit AI Integrations (OpenAI-compatible chat-completions endpoint; model gpt-5.4-mini)	Optional live invoice OCR: sends extracted PDF text or invoice images to return structured invoice fields. The packaged launcher defaults to cached synthetic fixtures, so no AI call occurs in the default path.	Paid usage via Replit credits when live OCR is used (per app UI); not identified as an SDC sponsor resource.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit platform / deployment configuration	Project platform configuration; .replit declares an Autoscale deployment target. The supplied run script launches local Vite and Express development servers	Autoscale target declared; actual plan, region, deployment URL and persistent-volume configuration not provided.
Local Node.js + Vite development runtime	Runs API on localhost:8080 and web UI on localhost:5173; Vite proxies /api to the API.	Local development; no cloud region/plan.
SQLite via better-sqlite3	Local invoice/activity/settings databases created on first run; no managed database service is configured in this demo launcher.	Local API filesystem; no cloud database plan/region.
Local private upload directory	Stores encrypted invoice-file bytes on the API server filesystem; no object-storage provider is configured.	Local filesystem; retention/region not specified.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-5.4-mini (exact model string in source)	Called through Replit AI Integrations using an OpenAI-compatible chat-completions client.	Pre-trained inference only; no fine-tuning or self-training pipeline identified.	Provider/Replit AI terms apply; model license, retention terms and endpoint region are not specified in the archive. Team should verify current terms.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic invoice extraction fixtures (6 sample invoices)	Billguard-Submission-Demo/server/data/extractions.json.	Synthetic	Team-created	N
Synthetic demo users and supplier records (3 simulated users; 5 suppliers)	Billguard-Submission-Demo/server/hub-mock/index.ts.	Synthetic	Team-created	N
Automated test and evaluation inputs	API/OCR/workflow/security tests and docs/verification.md	Synthetic	Team-created test material;	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Invoice files submitted by an operator (runtime input; not a training dataset)	Uploaded by a user at runtime; not a bundled external dataset. Demo README instructs users to use synthetic samples only.	Private / user-provided	Uploader must have rights to submit the document.	Y — may contain personal, supplier, tax or financial information; real data is prohibited in the demo.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit	Coding	Major
Claude	Coding and consulting	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team run the API tests and used the pytest environment to run the tests for the web application.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Credits	Used to provide higher level AI input during the process of coding and debugging
Runpod	Credits	Used for severless GPU to complete OCR function in Billguard

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

We use sponsor-provided Replit credits and RunPod credits. Replit credits were used in fine tuning the code behind the project as well as helping debug syntax and logical errors. $25 worth credits will be used for a serverless GPU to compute the OCR function in our project. Both Replit and Runpod played a large role in our development, and Runpod will further be core to the functionality of Billguard's leading features.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Billguard processes invoices uploaded by staff which will contain supplier names and TIN, invoice numbers, dates and other data. We currently don't have a database connected and are testing the application with dummy data. Encryption for safety and privacy is a feature that we are working on developing in the (hopefully) near future.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

OCR may misread, omit, or infer fields. However as the model is dependent on cross referencing of the data by the accountant as well as requiring a final approval by the owner, the data generated by the OCR is human verified.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature 

Date 24/10/2026